

Angular 19 Full Project Guide

This document contains the complete Angular 19 frontend setup for Student Teacher Portal with Login, Register, JWT Token Storage, Dashboard, and Logout functionality.

Step 1 - Create Angular Project

```
ng new student-teacher-ui
cd student-teacher-ui
```

Step 2 - Install Bootstrap

```
npm install bootstrap
```

Step 3 - Add Bootstrap in angular.json

```
"styles": [
  "node_modules/bootstrap/dist/css/bootstrap.min.css",
  "src/styles.css"
]
```

Step 4 - Generate Components

```
ng generate component pages/landing
ng generate component auth/login
ng generate component auth/register
ng generate component dashboards/student-dashboard
ng generate component dashboards/teacher-dashboard
ng generate component shared/navbar
```

Step 5 - Generate Service

```
ng generate service services/auth
```

Step 6 - app.routes.ts

```
import { Routes } from '@angular/router';

export const routes: Routes = [
  {
    path: '',
    component: LandingComponent
  },
  {
    path: 'login',
    component: LoginComponent
  },
  {
    path: 'register',
    component: RegisterComponent
  },
  {
    path: 'student-dashboard',
    component: StudentDashboardComponent
  },
  {
    path: 'teacher-dashboard',
    component: TeacherDashboardComponent
  }
];
```

Step 7 - Environment File

```
export const environment = {
  apiUrl: 'http://127.0.0.1:8000/api'
};
```

Step 8 - Auth Service

```
import { Injectable } from '@angular/core';
import { HttpClient } from '@angular/common/http';

@Injectable({
  providedIn: 'root'
})
export class AuthService {

  apiUrl = 'http://127.0.0.1:8000/api';

  constructor(private http: HttpClient) {
  }

  register(data: any)
  {
    return this.http.post(
      `${this.apiUrl}/register`,
      data
    );
  }

  login(data: any)
  {
    return this.http.post(
      `${this.apiUrl}/login`,
      data
    );
  }
}
```

Step 9 - Generate Auth Guard

```
ng generate guard guards/auth
```

auth.guard.ts

```
import { CanActivateFn, Router } from '@angular/router';
import { inject } from '@angular/core';

export const authGuard: CanActivateFn = (route, state) => {

  const router = inject(Router);

  const token = localStorage.getItem('token');

  if (token) {
    return true;
  }

  router.navigate(['/login']);

  return false;
};
```

Step 10 - Landing Page

landing.component.ts

```
import { Component } from '@angular/core';
import { RouterLink } from '@angular/router';
import { NavbarComponent } from '../../../shared/navbar/navbar.component';

@Component({
  selector: 'app-landing',
  imports: [NavbarComponent, RouterLink],
  templateUrl: './landing.component.html',
  styleUrls: ['./landing.component.scss']
})
export class LandingComponent {

}
```

landing.component.html

```
<app-navbar></app-navbar>

<div class="main-section">

  <div class="container">

    <div class="row align-items-center min-vh-100">

      <div class="col-md-6 text-center text-md-start">

        <h1 class="main-title">
          Student Teacher Portal
        </h1>

        <p class="sub-text mt-3">
          Simple learning management system using
          Angular 19 + Laravel 12 + JWT Authentication
        </p>

        <div class="mt-4">
          <a routerLink="/register" class="btn btn-primary btn-lg me-3">
            Register
          </a>

          <a routerLink="/login" class="btn btn-outline-dark btn-lg">
            Login
          </a>
        </div>

      </div>

      <div class="col-md-6 text-center">

        

      </div>

    </div>

  </div>

</div>
```

Step 11 - Login Page

login.component.ts

```
import { Component } from '@angular/core';

import { FormsModule } from '@angular/forms';

import { Router, RouterLink } from '@angular/router';

import { NavbarComponent } from '../../../shared/navbar/navbar.component';

import { AuthService } from '../../../services/auth.service';

@Component({
  selector: 'app-login',
  imports: [FormsModule, RouterLink, NavbarComponent],
  templateUrl: './login.component.html',
  styleUrls: ['./login.component.scss']
})
export class LoginComponent {

  formData = {

    vch_email: '',

    vch_password: ''

  };

  constructor( private authService: AuthService, private router: Router) {

  }

  login() {

    // console.log(this.formData); return false;

    this.authService.login(this.formData).subscribe({

      next: (response) => {

        // console.log(response); return false;

        /*
          Store JWT Token
        */

        localStorage.setItem(
          'jwt_token',
          response.jwt_token
        );

        /*
          Store User Data
        */

        localStorage.setItem(
          'user',
          JSON.stringify(response.user)
        );

        alert(response.message);
```



```

                name="vch_password"
                [(ngModel)]="formData.vch_password"
            >
        </div>

        <button class="btn btn-success w-100">
            Login
        </button>

    </form>

</div>
</div>
</div>
</div>
</div>

```

Step 12 - Register Page

register.component.ts

```

import { Component } from '@angular/core';

import { FormsModule } from '@angular/forms';

import { RouterLink } from '@angular/router';

import { NavbarComponent } from '../shared/navbar/navbar.component';

import { AuthService } from '../services/auth.service';

import { Router } from '@angular/router';

@Component({
  selector: 'app-register',
  imports: [FormsModule, RouterLink, NavbarComponent],
  templateUrl: './register.component.html',
  styleUrls: ['./register.component.scss']
})
export class RegisterComponent {

  formData = {

    vch_name: '',

    vch_email: '',

    vch_password: '',

    int_role: ''

  };

  constructor(private authService: AuthService, private router: Router) {

  }

  register() {

    // console.log(this.formData); return false;
  }
}

```

```

    this.authService.register(this.formData).subscribe({

      next: (response) => {

        console.log(response);

        alert(response.message);

        this.router.navigate(['/login']);

      },

      error: (error) => {

        console.log(error);

        alert('Something went wrong');

      }

    });
  }
}

```

register.component.html

```

<app-navbar></app-navbar>

<div class="container mt-5">
  <div class="row justify-content-center">

    <div class="col-md-5">

      <div class="card shadow-lg border-0">
        <div class="card-body p-4">

          <h2 class="text-center mb-4">
            Register
          </h2>

          <form (ngSubmit)="register()">

            <div class="mb-3">
              <label>Name</label>

              <input
                type="text"
                class="form-control"
                placeholder="Enter name"
                name="vch_name"
                [(ngModel)]="formData.vch_name"
              >
            </div>

            <div class="mb-3">
              <label>Email</label>

              <input
                type="email"
                class="form-control"
                placeholder="Enter email"
                name="vch_email"
                [(ngModel)]="formData.vch_email"
              >
            </div>

```

```

        <div class="mb-3">
            <label>Password</label>

            <input
                type="password"
                class="form-control"
                placeholder="Enter password"
                name="vch_password"
                [(ngModel)]="formData.vch_password"
            >
        </div>

        <div class="mb-3">
            <label>Select Role</label>

            <select
                class="form-select"
                name="int_role"
                [(ngModel)]="formData.int_role"
            >
                <option value="">
                    Choose Role
                </option>

                <option value="1">
                    Teacher
                </option>

                <option value="2">
                    Student
                </option>
            </select>
        </div>

        <button class="btn btn-primary w-100">
            Register
        </button>

    </form>

</div>
</div>
</div>
</div>
</div>

```

Step 13 - Student Dashboard

student-dashboard.component.ts

```

import { Component } from '@angular/core';

import { Router } from '@angular/router';

@Component({
  selector: 'app-student-dashboard',
  standalone: true,
  imports: [],
  templateUrl: './student-dashboard.component.html',
  styleUrls: ['./student-dashboard.component.css']
})

```



```

export class StudentDashboardComponent {

  user: any;

  constructor(private router: Router)
  {
    const token = localStorage.getItem('jwt_token');

    // If token missing redirect login
    if(!token)
    {
      this.router.navigate(['/login']);
    }

    const userData = localStorage.getItem('user');

    if(userData)
    {
      this.user = JSON.parse(userData);
    }

    // Prevent browser back after logout
    history.pushState(null, '', location.href);

    window.onpopstate = () => {

      history.pushState(null, '', location.href);

      const checkToken = localStorage.getItem('jwt_token');

      if(!checkToken)
      {
        this.router.navigate(['/login']);
      }
    };
  }

  logout()
  {
    /*
      Clear Local Storage
    */

    localStorage.removeItem('jwt_token');

    localStorage.removeItem('user');

    /*
      Redirect Login
    */

    this.router.navigate(['/login']);
  }
}

```

student-dashboard.component.html

```

<div class="container mt-5">

  <div class="card shadow-lg border-0">

    <div class="card-body p-5 text-center">

      <h1 class="mb-4 text-primary">

```

```

        Student Dashboard
    </h1>

    <h3>
        Welcome,
        {{ user?.vch_name }}
    </h3>

    <p class="mt-3">
        Email:
        {{ user?.vch_email }}
    </p>

    <p>
        Role: {{ user?.int_role == 1 ? 'Teacher' : 'Student' }}
    </p>

    <button
        class="btn btn-danger mt-3"
        (click)="logout()"
    >
        Logout
    </button>

</div>

</div>

</div>

```

Step 14 - Teacher Dashboard

teacher-dashboard.component.ts

```

import { Component } from '@angular/core';

import { Router } from '@angular/router';

@Component({
  selector: 'app-teacher-dashboard',
  standalone: true,
  imports: [],
  templateUrl: './teacher-dashboard.component.html',
  styleUrls: ['./teacher-dashboard.component.css']
})

export class TeacherDashboardComponent {

  user: any;

  constructor(private router: Router)
  {
    const token = localStorage.getItem('jwt_token');

    // If token not found redirect login
    if(!token)
    {
      this.router.navigate(['/login']);
    }

    const userData = localStorage.getItem('user');

    if(userData)
    {

```

```

        this.user = JSON.parse(userData);
    }

    // Prevent browser back after logout
    history.pushState(null, '', location.href);

    window.onpopstate = () => {

        history.pushState(null, '', location.href);

        const checkToken = localStorage.getItem('jwt_token');

        if(!checkToken)
        {
            this.router.navigate(['/login']);
        }
    };
}

logout()
{
    localStorage.removeItem('jwt_token');

    localStorage.removeItem('user');

    this.router.navigate(['/login']);
}
}

```

teacher-dashboard.component.html

```

<div class="container mt-5">

    <div class="card shadow-lg border-0">

        <div class="card-body p-5 text-center">

            <h1 class="mb-4 text-success">
                Teacher Dashboard
            </h1>

            <h3>
                Welcome,
                {{ user?.vch_name }}
            </h3>

            <p class="mt-3">
                Email:
                {{ user?.vch_email }}
            </p>

            <p>
                Role: {{ user?.int_role == 1 ? 'Teacher' : 'Student' }}
            </p>

            <button
                class="btn btn-danger mt-3"
                (click)="logout()"
            >
                Logout
            </button>

        </div>

    </div>

```

```
</div>
```

```
</div>
```

Step 15 - Run Angular Project

```
ng serve
```